The Equation of a Hyperplane

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Dot product.
- **Leads to:** Perceptron trick; Support vector machines; Perceptron.

1. Overview

Key point: One equation, $w^\mathsf{T}x + w_0 = 0$, describes a line in 2D, a **plane** (G-1502) in 3D and a hyperplane in any number of dimensions; w is perpendicular to it and w_0 moves it away from the origin.

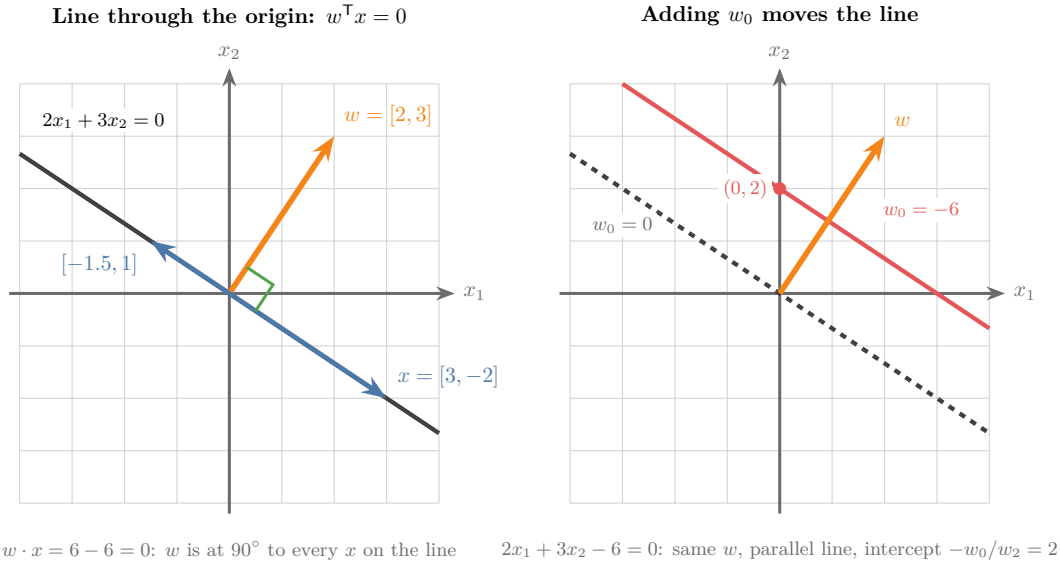


Figure 1: Left: a line through the origin and the vector w perpendicular to it. Right: adding w_0 moves the line without turning it

Figure 1 shows the whole result in 2D. On the left, every vector x on the line makes a right angle with w, so their dot product is zero. On the right, adding a number w_0 slides the line to a parallel position. This Note builds that equation step by step from the school equation of a line, and then reads off what w and w_0 mean.

The symbols in the equation:

- x is a point, written as a **vector** (a list of coordinates, such as $[3, 0]$);
- w is a vector of coefficients, such as $[2, 3]$;

- $w^\mathsf{T}x$ is the **dot product** of the two: multiply matching entries and add. For these two vectors:

$$w^\mathsf{T}x = 2 \times 3 + 3 \times 0$$

$$w^\mathsf{T}x = 6$$

- w_0 is one number (a single value).

The dot product and its geometric form are taught in the geometric meaning of the dot product.

2. Why ML needs hyperplanes

> **Key point:** A line in 2D becomes a plane in 3D and a hyperplane beyond; models that classify with a straight boundary need its equation in any dimension.

In 2D a flat boundary is a line; in 3D it is a plane. In 4D, 5D or n-D the equivalent is a **hyperplane** (G-911; see more inputs, a hyperplane).

ML data is very often high-dimensional. Each **feature** (G-772; an input variable, one column of the data table) adds one dimension. Even the iris toy dataset has 4 features, and MNIST, the handwritten digits dataset, has 784 (one per pixel of a 28 x 28 image). Classifiers that separate classes with a straight boundary, such as **logistic regression** (G-1120) and SVM, therefore work with hyperplanes. To use them, and to code such algorithms ourselves, we need one equation that works in every dimension.

Figure 2 shows one such boundary on real data. With three features of the iris flowers (sepal length, petal length and petal width), a linear SVM fitted with scikit-learn finds a **plane** (G-1502) with every setosa flower on one side and every versicolor flower on the other. With all four features the same boundary would be a hyperplane, which we can no longer draw but can still write down.

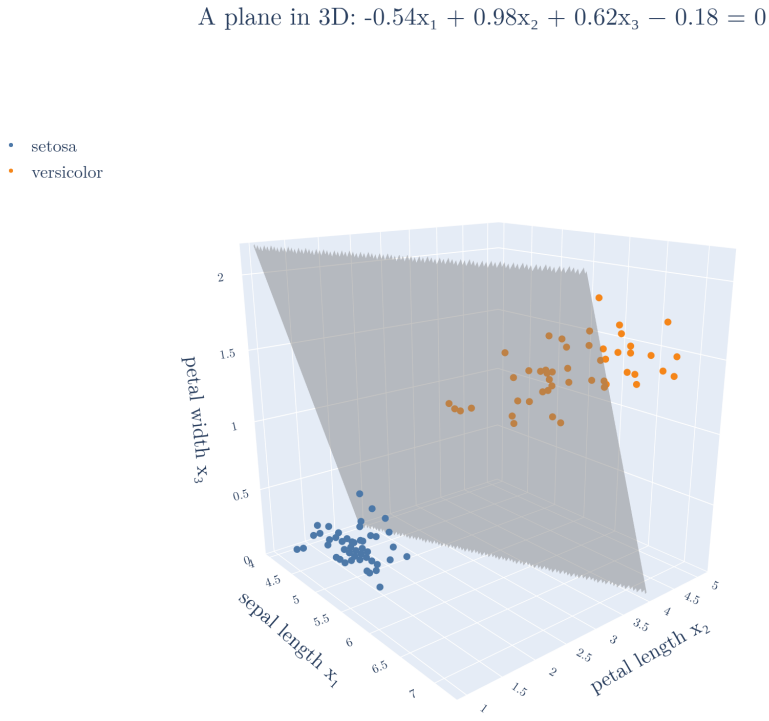


Figure 2: Iris setosa (blue) and versicolor (orange) in three features, separated by the plane $-0.54\mathrm{x}_1 + 0.98\mathrm{x}_2 + 0.62\mathrm{x}_3 - 0.18 = 0$ that a linear SVM (scikit-learn LinearSVC) found.

3. From a line to a hyperplane

> **Key point:** Rename the general form of a line as $w_1x_1 + w_2x_2 + w_0 = 0$; each extra dimension adds one more term w_ix_i.

3.1 The general form of a line

> **Key point:** $y = mx + b$ and $ax + by + c = 0$ describe the same line; the general form is the one that grows into higher dimensions.

In school a line is $y = mx + b$, with **slope** (G-1823) m and **intercept** (G-960) b. The same line has a general form, $ax + by + c = 0$, used in the general form of a line.

Solving the general form for y links the two:

$$y = -\frac{a}{b}\,x - \frac{c}{b}$$

So the slope and the intercept are:

$$m = -\frac{a}{b}$$

$$\text{intercept} = -\frac{c}{b}$$

(the b of the general form is a different number from the intercept b of $y = mx + b$). We use the general form, because it treats every axis the same way.

3.2 Renaming for n dimensions

> **Key point:** Call the axes $x_1, x_2, \ldots$ and the coefficients $w_1, w_2, \ldots$, with w_0 for the constant.

Letters x, y, z run out after three dimensions, so we number the axes instead: $x_1, x_2, x_3, \ldots$ We number the coefficients the same way, $w_1, w_2, \ldots$, and call the constant term w_0. Then:

- **Line (2D):**

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

- **Plane (3D):**

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$$

- **Hyperplane (n-D):**

$$w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0 = 0$$

Each new dimension adds one term. With numbers, the line $2x + 3y - 6 = 0$ is renamed:

$$2x_1 + 3x_2 - 6 = 0$$

Here $w_1 = 2$, $w_2 = 3$ and $w_0 = -6$. Figure 3 (left) draws it in the school form, with a **slope** (G-1823) and an intercept:

$$\text{slope} = -\frac{a}{b} = -\frac{2}{3}$$

$$\text{intercept} = -\frac{c}{b} = 2$$

The right panel plugs points into the equation, which section 4 writes as a dot product.

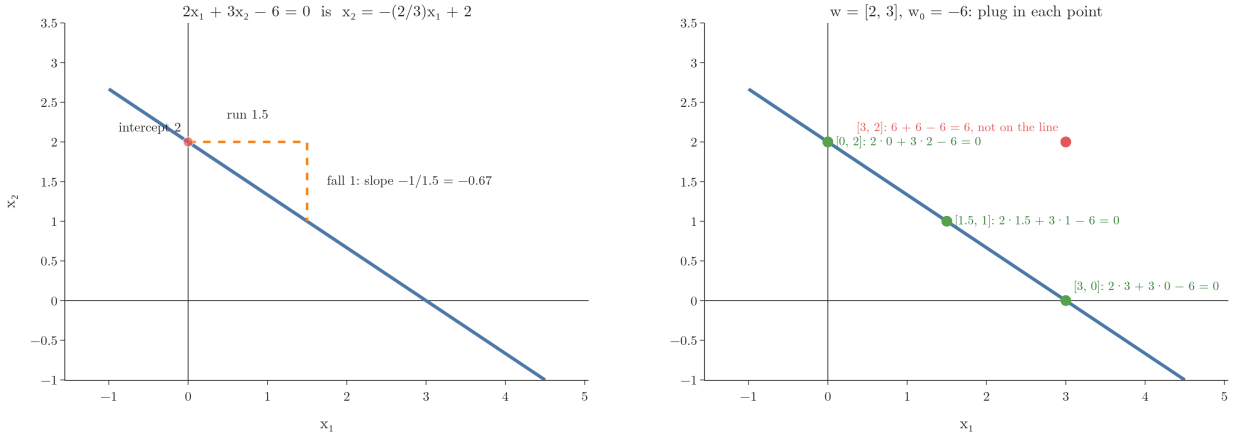


Figure 3: The line $2x_1 + 3x_2 - 6 = 0$. Left: the same line as $x_2 = -(2/3)x_1 + 2$, with its slope triangle and intercept 2. Right: [3, 0], [0, 2] and [1.5, 1] give 0 and lie on the line; [3, 2] gives 6 and does not.

4. The vector form

Key point: The sum of the products $w_i x_i$ is a dot product, so the hyperplane is written in vector form:

$$w \cdot x + w_0 = 0$$

$$w^{\mathsf{T}} x + w_0 = 0$$

Look at the sum of the first part of the equation:

$$w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

This sum multiplies matching components and adds them, which is exactly a dot product. Collect the coefficients into one vector and the coordinates into another:

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Both are column vectors, the default. Writing the dot product as a row times a column, $w \cdot x = w^\mathsf{T} x$, gives the equation of a hyperplane:

1. **In words:** the dot product of the coefficient vector with any point on the hyperplane, plus the constant, is zero.
2. **Formula:**
$$w^\mathsf{T} x + w_0 = 0$$
3. **Example:** for $w = [2, 3]$ and $w_0 = -6$, the point $x = [3, 0]$ gives
$$w^\mathsf{T} x + w_0 = 2 \times 3 + 3 \times 0 - 6$$
$$w^\mathsf{T} x + w_0 = 6 + 0 - 6 = 0$$
so $[3, 0]$ lies on the line (Figure 3, right).

This one equation is valid in 2D, 3D and n-D: only the number of components of w and x changes. The equation is the same $w \cdot u + b$ that SVM uses in the decision rule, with $b = w_0$.

5. What w_0 means

> **Key point:** w_0 shifts the hyperplane away from the origin; $w_0 = 0$ exactly when the hyperplane passes through the origin.

To see what w_0 does, go back to 2D, where the line is both $x_2 = mx_1 + b$ and $w_1 x_1 + w_2 x_2 + w_0 = 0$. Section 3.1 matched the **intercepts** (G-960):

$$b = -\frac{w_0}{w_2}$$

Now slide the line down until it passes through the origin. Its intercept b becomes 0, and since $w_2 \neq 0$, that forces $w_0 = 0$. So:

- A line, **plane** (G-1502) or hyperplane passes through the origin exactly when $w_0 = 0$.
- A non-zero w_0 moves it parallel to itself, away from the origin. In Figure 1 (right), this line passes through the origin:
$$2x_1 + 3x_2 = 0$$
This line, with $w_0 = -6$, is the same line shifted:
$$2x_1 + 3x_2 - 6 = 0$$
It crosses the x_2 axis at the intercept:
$$b = -\frac{w_0}{w_2} = -\frac{-6}{3} = 2$$

Figure 4 turns both knobs on real data: the iris setosa and versicolor flowers, plotted by petal length x_1 and petal width x_2 (in cm). Watch the line while w_0 changes: it slides without turning, and the arrow w keeps its direction. Turning w instead tilts the line, which stays at 90° to w (Section 6). The shaded side is where $w^\mathsf{T}x + w_0 > 0$, the side w points to (the sign rule at the end of Section 6). With $w = [1, 1]$ and $w_0 = -3.2$, all 50 versicolor flowers fall on the shaded side and all 50 setosa flowers on the other.

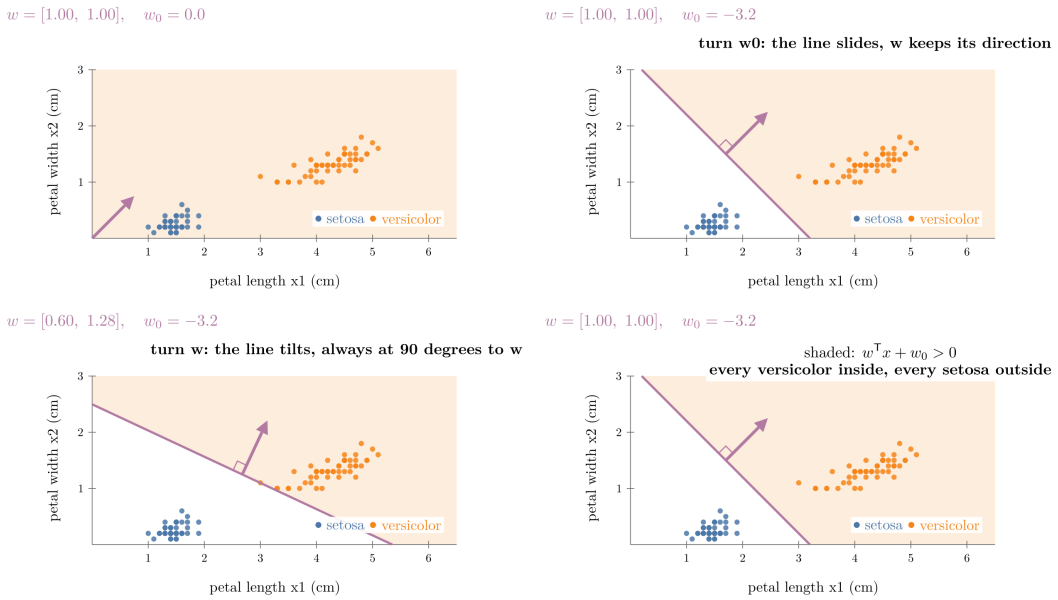


Figure 4: The two knobs of $w^\mathsf{T}x + w_0 = 0$ on iris setosa (blue) and versicolor (orange): changing w_0 slides the line, turning w tilts it; the shaded side is where $w^\mathsf{T}x + w_0 > 0$

So we often take the simplifying assumption that the hyperplane passes through the origin. The equation then shrinks to

$$w^\mathsf{T}x = 0$$

In 2D that is:

$$w_1x_1 + w_2x_2 = 0$$

In 4D it is:

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 = 0$$

> **Extra:** The assumption costs nothing, because of a standard trick. Add a constant component 1 to every point, and put w_0 into w:
>
> $$x = [1, x_1, \ldots, x_n]$$

$$w = [w_0, w_1, \ldots, w_n]$$

Then the product contains the shift:

$$w^\mathsf{T} x = w_0 + w_1 x_1 + \cdots + w_n x_n$$

So the equation $w^\mathsf{T} x = 0$ already contains w_0. The constant component is the column of ones in the matrix form of stacking the equations in multiple linear regression.

6. What w means: the normal vector

Key point: $w^\mathsf{T} x = 0$ says w makes a 90° angle with every point x on the hyperplane, so w is perpendicular to the hyperplane.

6.1 The argument

Key point: A dot product of non-zero vectors is 0 only at 90°.

For a hyperplane through the origin, every point x on it satisfies $w^\mathsf{T} x = 0$. Write the dot product in its geometric form:

$$w^\mathsf{T} x = w \cdot x = \|w\| \, \|x\| \cos\theta = 0$$

For non-zero w and x, the lengths are positive, so $\cos\theta = 0$ and $\theta = 90°$. Each point x on the hyperplane is a vector lying in the hyperplane, and w is at a right angle to every one of them. So w is perpendicular to the hyperplane itself. A vector perpendicular to a line, **plane** (G-1502) or hyperplane is called its **normal vector** (G-1346).

With numbers, take the line in Figure 1 (left):

$$2x_1 + 3x_2 = 0$$

Here $w = [2, 3]$, and the points $[3, -2]$ and $[-1.5, 1]$ lie on the line. Both give a dot product of zero:

$$2 \times 3 + 3 \times (-2) = 0$$

$$2 \times (-1.5) + 3 \times 1 = 0$$

6.2 The same in 3D

> **Key point:** The coefficients of a **plane** (G-1502)'s equation, read as a vector, stick straight out of the plane.

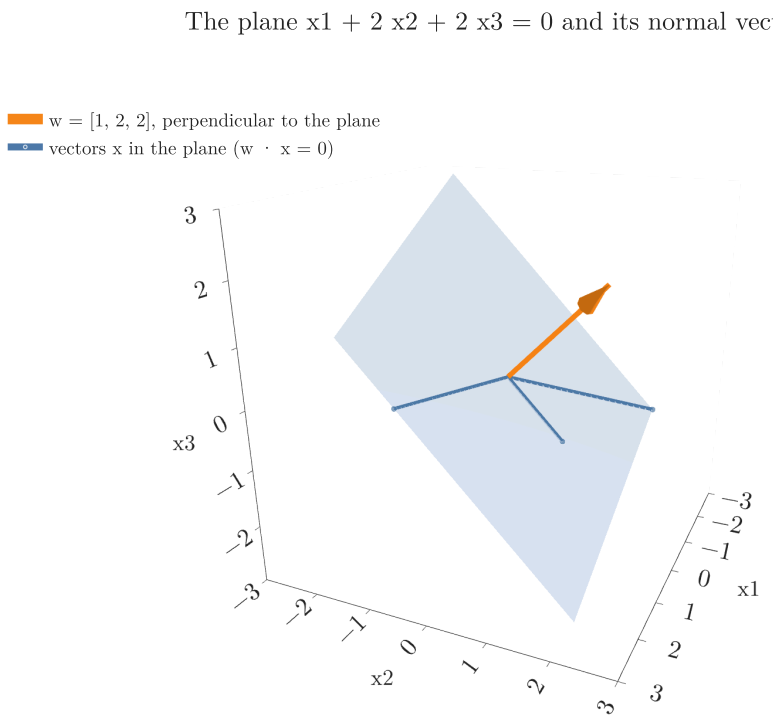


Figure 5: The plane x1 + 2x2 + 2x3 = 0 and its normal vector w = [1, 2, 2]

Figure 5 shows this plane:

$$x_1 + 2x_2 + 2x_3 = 0$$

Its coefficients form $w = [1, 2, 2]$, drawn in orange. The three blue vectors lie in the plane, for example $[2, -1, 0]$:

$$1 \times 2 + 2 \times (-1) + 2 \times 0 = 0$$

The orange w stands at 90° to all of them.

So reading a hyperplane's equation tells us its direction at once: the coefficients are the **normal vector** (G-1346). In n-D we cannot draw it, but the argument of Section 6.1 never used the number of dimensions.

Python: Checking that points lie on a hyperplane is one dot product per point.

```python
import numpy as np

w = np.array([1, 2, 2])
X = np.array([[2, -1, 0], [0, 1, -1], [-2, 2, -1]])
# dot product of every row of X with w
X @ w    # array([0, 0, 0]): all on the plane
```

6.3 A plane from a point and a normal vector

Key point: A point x_0 and a **normal vector** (G-1346) w fix a **plane** (G-1502): it holds every x whose arrow from x_0 is at right angles to w. Expanding gives the usual equation, with the constant w_0 set by the point, so w is the normal vector for any w_0.

Sections 6.1 and 6.2 assumed $w_0 = 0$. The same result holds for a hyperplane anywhere in space, and Figure 6 shows why in four steps.

1. **One point is not enough.** Think of the plane as a sheet of cardboard pinned at one point, $x_0 = (1, 1, 1)$. The sheet can still pivot about the pin, so one point leaves the plane open.
2. **A normal vector fixes the plane.** Choose the direction that must stand at 90° to the sheet, here $w = [1, 2, 2]$. Only one position of the sheet fits.
3. **Arrows inside the plane.** Take any other point x on the plane. The points x_0 and x are vectors from the origin, and they do not lie in the plane. Their difference $x - x_0$ does: it is the arrow from the pin to x, drawn on the cardboard. So w is at 90° to it, and their dot product is 0 (section 6.1).
4. **The constant slides the plane.** Changing w_0 moves the plane along w without turning it.

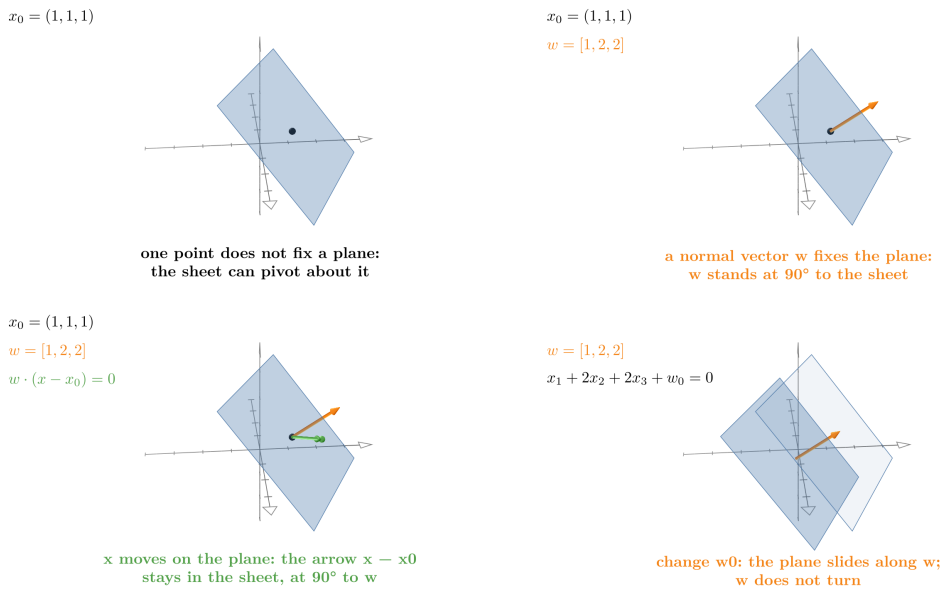


Figure 6: A plane from the point x0 = (1, 1, 1) and the normal vector w = [1, 2, 2]. The sheet can pivot about one point; the normal vector fixes it; the arrow from x0 to any point x of the plane stays at 90° to w; changing w0 slides the plane along w. Idea after Khan Academy, "Defining a plane in R3 with a point and normal vector".

Worked example. With $w = [1, 2, 2]$ and $x_0 = (1, 1, 1)$, a point $x = (x_1, x_2, x_3)$ is on the plane when

$$w \cdot (x - x_0) = 0$$

$$1(x_1 - 1) + 2(x_2 - 1) + 2(x_3 - 1) = 0$$

Multiplying out, one term per line:

$$1(x_1 - 1) = x_1 - 1$$

$$2(x_2 - 1) = 2x_2 - 2$$

$$2(x_3 - 1) = 2x_3 - 2$$

Adding the three terms:

$$x_1 + 2x_2 + 2x_3 - 1 - 2 - 2 = 0$$

$$x_1 + 2x_2 + 2x_3 - 5 = 0$$

The coefficients are again w, and the constant is $w_0 = -5$. The point $(3, 1, 0)$ lies on this plane:

$$3 + 2 + 0 - 5 = 0$$

The formal version. For any number of dimensions,

$$w^\mathsf{T}(x - x_0) = 0 \quad \Longleftrightarrow \quad w^\mathsf{T}x + w_0 = 0$$

$$w_0 = -w^\mathsf{T}x_0$$

Read from right to left, the same steps show that w stays perpendicular when $w_0 \neq 0$. Take any two points x and y on the hyperplane:

$$w^\mathsf{T}x + w_0 = 0$$

$$w^\mathsf{T}y + w_0 = 0$$

Subtracting the second from the first:

$$w^\mathsf{T}(x - y) = 0$$

The vector $x - y$ runs along the hyperplane, so w is perpendicular to every direction in it.

All hyperplanes with the same w and different w_0 therefore share one normal vector and are parallel. The shared normal vector is why the two lines in Figure 1 (right) are parallel, and why the plane of Figure 5 is parallel to the plane of this example (both have $w = [1, 2, 2]$).

> **Extra:** For a point off the hyperplane, $w^\mathsf{T}x + w_0$ is not zero, and its sign says which side the point is on: positive on the side w points to, negative on the other. To see this, start at a point p on the hyperplane and step a distance t along w: $x = p + t\,w/\|w\|$. Then, one step per line:
>
> $$w^\mathsf{T}x + w_0 = (w^\mathsf{T}p + w_0) + t\,\frac{w^\mathsf{T}w}{\|w\|}$$
>
> $$= 0 + t\,\frac{\|w\|^2}{\|w\|} = t\|w\|$$
>
> which has the sign of t. This sign rule is the side test of **positive and negative sides** (G-1529) in the perceptron trick and the decision rule of the SVM.

7. Summary

Object	Equation	Normal vector
Line (2D)	$w_1x_1 + w_2x_2 + w_0 = 0$	$[w_1, w_2]$
Plane (3D)	$w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$	$[w_1, w_2, w_3]$
Hyperplane (n-D)	$w^\mathsf{T}x + w_0 = 0$	w
Through the origin	$w^\mathsf{T}x = 0$	w

- The general form of a line grows into a hyperplane by adding one term per dimension; the sum of terms is a dot product, so one vector equation covers every dimension.
- w_0 shifts the hyperplane; a zero w_0 means it passes through the origin, because the **intercept** (G-960) is 0 exactly when w_0 is 0.
- w is the **normal vector** (G-1346): perpendicular to the hyperplane, because a zero dot product means a 90° angle.
- A point x_0 and a normal vector w give the hyperplane through that point; its constant is minus the dot product of w and x_0, so w stays the normal vector for any w_0, and hyperplanes with the same w are parallel.
- So one equation, $w^\mathsf{T}x + w_0 = 0$, describes a straight boundary in any number of dimensions, which is what classifiers such as **logistic regression** (G-1120) and SVM need.

8. Sources

Built from

- CampusX, "Equation of a Hyper-plane in N dimensions", YouTube, https://www.youtube.com/watch?v
b8AgdVA
- CampusX, "Supercharge Your ML Journey: Mastering Vectors in Linear Algebra - Part 1", YouTube, https://www.youtube.com/watch?v=mQewAJb8oJ8
- Khan Academy, "Defining a plane in R3 with a point and normal vector | Linear Algebra | Khan Academy", YouTube, https://www.youtube.com/watch?v=UJxgcVaNTqY
- Khan Academy, "Normal vector from plane equation | Vectors and spaces | Linear Algebra | Khan Academy", YouTube, https://www.youtube.com/watch?v=gw-4wltP5tY

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Normal vector (G-1346)	A vector perpendicular to a line, plane or hyperplane; for $w^\mathsf{T}x + w_0 = 0$ it is w.

Term	Meaning
w_0 (G-38)	The fixed number in the equation of a line, plane or hyperplane (the constant term). It shifts the hyperplane away from the origin, and is 0 when the hyperplane passes through the origin.
Equation of a hyperplane (G-703)	One equation for a flat boundary in any number of dimensions, a line in 2D, a plane in 3D, a hyperplane beyond: $w^\mathsf{T}x + w_0 = 0$.
Plane (G-1502)	A flat surface in 3D; the model for two input columns.
Hyperplane (G-911)	A flat surface in more than three dimensions; the model for three or more input columns.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Logistic regression (G-1120)	A classification algorithm that passes a weighted sum of the inputs through the sigmoid to give the probability of a class; the line (or plane) where that probability is 0.5 is the boundary it learns between the classes.
Slope (G-1823)	How much the output changes for one unit of change in the input; m in $y = mx + b$.
Intercept (G-960)	The line's value when the input is 0; b in $y = mx + b$.
Positive and negative side (G-1529)	The two halves of the plane on either side of a line $Ax + By + C = 0$: where $Ax + By + C$ is above 0 and where it is below 0; the perceptron predicts one class on each side.